Business Entity Resolution Across Three Noisy Sources: A Precision-Bounded, Single-Stage Matching System

Amazon ML Challenge 2026 — Business Entity Resolution

Team Grokking
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Abstract. Three independent sources describe the same businesses with noisy, inconsistent names and addresses and share no common identifier. For every record of the clean reference source S1, the task is to find every record in the noisy sources S2 and S3 that refers to the same business, scored by macro $F_{0.5}$ per S1 entity — a metric that weights precision twice as heavily as recall and gives full credit for correctly predicting no match. Our system retrieves candidates with a deduplicated union of typed-token TF-IDF retrieval and exact structured keys, scores each (S1, candidate) pair with a single LightGBM classifier, and assigns every downstream record to at most one S1 entity via a global, calibrated acceptance threshold with an explicit NULL option. Every component that reached the final system was admitted only after a forensic and ablation process that rejected several more complex alternatives — a second, entity-level rescoring stage, character n-gram retrieval, and hard-negative reweighting — because they either leaked information or did not reproducibly improve the audited macro $F_{0.5}$. The submitted configuration scored **98.4/100** on the final official leaderboard (macro $F_{0.5} \approx 0.984$), placing the team in the **top 2.9%** (final rank $\approx$930) among a field of over 32,000 teams.

Keywords: entity resolution; record linkage; blocking; learning to match; gradient boosting; macro $F_{0.5}$.

1 Introduction

Entity resolution decides which records, drawn from sources that share no key, describe the same real-world object. In this challenge, three sources of business records each carry only a name, an address and a country label. Source 1 (S1) is a clean, deduplicated reference list; Sources 2 and 3 (S2, S3) contain noisy copies of S1 businesses, plus unrelated distractor records. The training data covers the United States and India; the test data adds a third, unlabeled country, France, which never appears in training and makes up roughly 15% of the test S1 population.

The standard recipe for this class of problem — normalise, block, score pairs, threshold — is a reasonable starting point, and it is what our system implements. What distinguishes our approach is not additional model capacity but a disciplined process: every candidate improvement was checked against an independently reproduced implementation of the official metric, tested for leakage, and, where feasible, confirmed on an independent random seed before being admitted into the final system. Several plausible extensions were tested and rejected on this basis.

1.1 Contributions

- A ground-truth and source-identifier consistency audit establishing that every downstream record belongs to at most one S1 entity in the supplied labels, which we then use as a structural constraint on the decision rule (Section 3).

- A two-channel candidate generation stage combining typed-token TF-IDF retrieval with exact structured-key retrieval, deduplicated into a single candidate union carrying a corrected, non-positional retrieval score (Section 4).
- A single LightGBM pairwise scorer together with a deterministic one-owner-per-target-plus-NULL assignment rule, calibrated on a held-out partition and evaluated on a disjoint, locked audit partition (Sections 5–6).
- A pre-registered ablation record, including the components that were tested and rejected — a second, entity-level rescoring stage, character-level retrieval, and hard-negative reweighting — each with its measured effect (Section 7).
- An honest accounting of the system's remaining risk surface, centred on the unlabeled France cohort and unverified full-scale runtime and memory behaviour (Section 9).

2 Problem and Metric

For an S1 entity e with true match set M_e and predicted set $\hat{M}_e$,

$$F_{0.5}(e) = (1+\beta^2)\frac{P_e R_e}{\beta^2 P_e + R_e} = \frac{1.25\,|\hat{M}_e \cap M_e|}{0.25\,|M_e| + |\hat{M}_e|}, \qquad \beta = 0.5,$$

with $F_{0.5}(e) = 1$ when $M_e = \hat{M}_e = \emptyset$. The reported score is the macro average of $F_{0.5}(e)$ over every S1 entity in the evaluation set, singletons included.

Two properties of this metric shaped every design decision in the system. First, because $\beta = 0.5$, a false merge costs roughly twice as much as a missed link, so an uncertain candidate should generally be left unassigned rather than accepted. Second, because the average is taken over entities rather than pairs, a wrong merge on a true singleton is exactly as damaging to the score as a wrong merge on an entity with several true matches — singletons are not a minor edge case but a first-class part of the objective, which is why correct NULL prediction is built into the assignment rule rather than left as a byproduct of thresholding.

Worked example. For a predicted set {S2-00047, S2-00193, S3-00812} against a true set {S2-00047, S3-00812}: $|\hat{M} \cap M| = 2$, $|M| = 2$, $|\hat{M}| = 3$, so $P = 2/3$, $R = 1$, and $F_{0.5} = 1.25 \cdot 2/(0.25 \cdot 2 + 3) \approx 0.714$. Our independent scorer reproduces the published formula on this and on further worked cases, including the treatment of empty prediction against empty truth as a score of 1.

3 Data and Integrity Audit

3.1 Dataset

Each source file carries an `entity_id` (prefixed S1-, S2- or S3-), a `business_name`, a `business_address` and a `country` label. The ground-truth file maps each S1 identifier to a comma-separated list of matching S2/S3 identifiers. Training data covers two countries (US, India); the test data covers three (US, India, France), with 1,732,544 S1 entities in total and 259,452 of them (about 15%) belonging to France, a country entirely absent from training. Because the test set introduces a label unseen at training time, country is treated throughout as an open string field rather than a fixed, one-hot category.

Documented noise patterns include name abbreviations and legal-suffix inconsistencies (Corp/Corporation, Pvt/Private, Ltd/Limited), DBA/trade names, punctuation differences, word-order transpositions and typos; and address abbreviations, missing components, landmark-based references, municipal numbering variation and component reordering.

3.2 Ground-truth and source-identifier audit

Before any modelling, we joined the full ground-truth file against both source tables to check the structural assumptions the rest of the system depends on. Table 1 summarises the result: every ground-truth identifier resolves against its source table, and — critically — no target identifier in the labels is claimed by more than one S1 entity.

Check	Result
Ground-truth S1 rows / unique S1 IDs	2,206,821 / 2,206,821
Ground-truth target mentions / unique target IDs	7,638,365 / 7,638,365
Ground-truth S1 IDs absent from source S1	0
Ground-truth target IDs absent from source S2/S3	0
Target IDs with multiple S1 owners	0
Duplicate source S1 IDs; duplicate S2/S3 target IDs	0; 0

Table 1: Ground-truth / source-identifier consistency audit.

This result licenses a structural simplification used throughout the rest of the system: matching can be framed as, for every downstream record, choosing at most one S1 owner (Section 6), rather than as an unconstrained many-to-many pairing problem.

4 System Architecture

The system is a five-stage cascade: normalisation, country partitioning, candidate generation, pairwise scoring, and assignment. Each stage exchanges records by identifier only, so any stage can be replaced independently.

4.1 Normalisation

Names and addresses pass through deterministic, reversible normalisation: case folding and punctuation handling, script transliteration, legal-form canonicalisation, and extraction of an address-number view. Original identifiers are retained end to end so that every prediction is traceable to its source records. No country-specific normalisation branch was introduced, since France has no labels against which such a branch could be validated.

4.2 Candidate generation

Two complementary retrieval channels are merged into one candidate set:

- **Typed-token retrieval.** Tokens are typed (name word, number, legal form, ...) and scored with a TF-IDF-style weighted retrieval score against each S1 entity's token bag.
- **Exact-key retrieval.** Deterministic keys derived from normalised name and address fields catch near-identical records that token retrieval alone can rank too low.

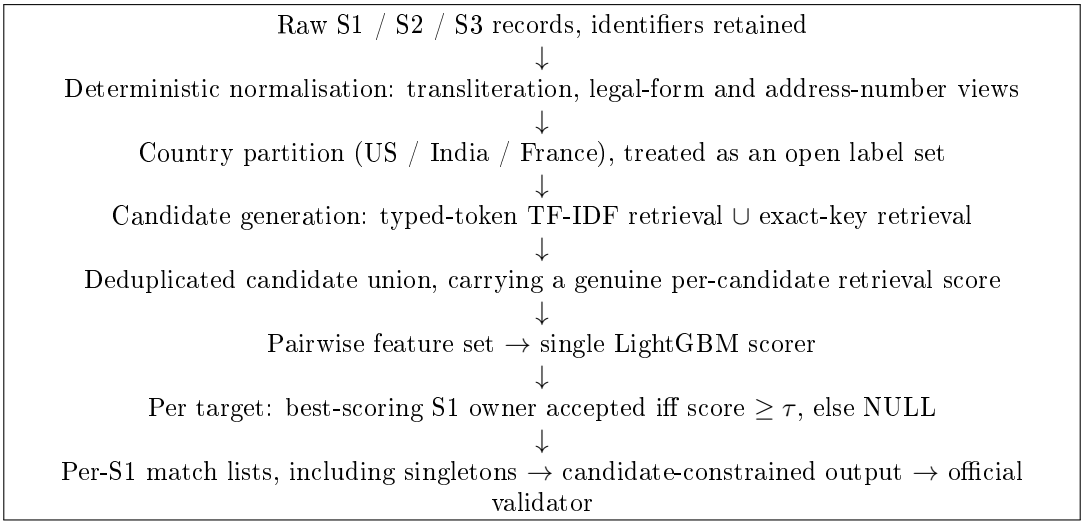


Figure 1: End-to-end pipeline of the submitted system. Only Stage 1 pairwise scoring is used; a candidate entity-level second stage was implemented, tested, and excluded (Section 7).

The union is deduplicated with deterministic tie-breaking, and each surviving candidate keeps its genuine typed-token retrieval score, exposed to the model as `tfidf_score`; exact-only candidates receive a value of zero for that feature. An earlier implementation had exposed the position of a candidate in the union as if it were a retrieval rank; this pseudo-rank was identified as non-evidentiary and replaced with the real score (Section 7). On audited samples, the merged candidate set retained close to 100% of true pairs (2,103/2,103 and 2,088/2,089 on two independent audit seeds).

5 Matching Model

A single LightGBM binary classifier scores every (S1, candidate) pair using the existing pairwise feature set, spanning:

- name similarity signals (token overlap, edit-distance style measures);
- address token and number agreement, including the address-number view;
- legal-form agreement between the S1 and candidate legal-form sets;
- transliteration-aware name comparison; and
- the corrected typed-token retrieval score, `tfidf_score`.

Training negatives are drawn from the natural retrieved-candidate distribution while excluding held-out positive target identifiers, and the calibration and locked audit partitions are separated at the S1 level so that no S1 entity's records appear in both. We deliberately kept the model to this single stage: a candidate second, entity-level rescoring stage was built and evaluated, but it is excluded from the submitted system (Section 7).

6 Decision and Assignment

Because the ground-truth audit (Table 1) shows that every downstream record belongs to at most one S1 entity, the assignment is made from the target's side rather than the S1 side. For each S2/S3 record r,

$$a(r) = \begin{cases} \arg\max_e p(r,e) & \text{if } \max_e p(r,e) \geq \tau, \\ \varnothing & \text{otherwise,} \end{cases}$$

with deterministic tie-breaking and τ selected on a calibration partition to maximise macro $F_{0.5}$, then frozen before scoring on the separate, locked audit partition. An S1 entity may own any number of targets; an S1 that owns none is emitted with an empty match list, which is exactly how the metric rewards correct singleton prediction (Section 2). We kept a single global threshold rather than a per-entity or per-country rule: with no France labels to calibrate against, a country-specific threshold could not be validated, and a global threshold avoids overfitting the small audited samples available for calibration.

7 Experimental Method and Ablation

All controlled experiments were run on ratio-matched, locked audit samples (around 600 S1 entities, drawn from a 6,000-S1 sampled index and sampled target corpus) and scored with our independently reproduced macro $F_{0.5}$ implementation. These are screening estimates, not full-scale competition estimates: they use a sampled index and target corpus and have limited coverage of rare-error strata and of the France cohort, which cannot be sampled at all. Table 2 summarises the components that were tested.

7.1 Why the second stage was rejected

A candidate second stage rescored surviving pairs using entity-level evidence (competing S1 owners, sibling records). Two audits found that earlier versions of this idea leaked information: first by fitting directly on validation rows, and later because in-sample second-stage predictions participated in the same owner competition they were trained on. After moving to strict three-fold, cross-fitted training on two independent seeds, one seed showed a small gain ($0.994138 \rightarrow 0.994617$, $+0.000479$) and the other showed a loss ($0.992561 \rightarrow 0.991372$, -0.001189); the two-seed mean effect is negative, so the second stage was excluded from the final system.

7.2 Why character retrieval was rejected

Adding character 3–5-gram retrieval as a third candidate-generation channel raised the oracle ceiling only marginally while adding 81,654 candidate edges for a single additional recovered true pair — a volume increase with essentially no matching benefit, and it was excluded on that basis.

Change	Macro $F_{0.5}$	Candidate effect	Decision
Genuine typed-token `tfidf_score` (seed A)	$0.9913 \rightarrow 0.9947$	unchanged	Keep
Genuine typed-token `tfidf_score` (seed B)	$0.9904 \rightarrow 0.9924$	unchanged	Keep
Typed + exact-key deduplication (seed A)	$0.9896 \rightarrow 0.9905$	no material growth	Keep
Strict out-of-fold second stage (seed A)	$0.9941 \rightarrow 0.9946$	unchanged	Insufficient alone
Strict out-of-fold second stage (seed B)	$0.9926 \rightarrow 0.9914$	unchanged	Reject — mean negative
Second-stage top-5 hard negatives	$0.9909 \rightarrow 0.9906$	unchanged	Reject
Character 3–5-gram retrieval	tiny oracle gain	+81,654 edges for +1 true pair	Reject

Table 2: Ablation record on locked audit samples. Each row changes exactly one component relative to its parent configuration.

7.3 Metric and pipeline correctness

The independently implemented macro $F_{0.5}$ scorer matched the published formula on the bounded audits, including correct treatment of singletons; no metric implementation defect was found. The full ground-truth/source-identifier audit (Table 1) likewise found no join, duplicate-owner, or duplicate-mention anomaly.

7.4 Components excluded from the core by design

Beyond the two rejected experiments above, the following were considered and left out of the submitted system without dedicated ablation runs, generally on cost, evidence, or rules grounds: a France-specific model or threshold (no labels to calibrate against), an additional address parser, embeddings and cross-encoders, graph-based propagation, assignment optimisation (e.g. Hungarian assignment), external data, and geocoding (the last two are prohibited by the competition rules regardless of measured effect).

8 Results

8.1 Pre-submission estimate

Locked audit samples produced Stage-1 macro $F_{0.5}$ between 0.9924 and 0.9947 (Table 2). Given the small sample size, the absence of any France labels, and no representative full-scale run, we treated this as an optimistic upper bound rather than a competition estimate, and worked with a more conservative pre-submission range of roughly 0.95–0.96 for the full evaluation.

8.2 Leaderboard outcome

The realised score exceeded our conservative pre-submission range, suggesting the unlabeled France cohort transferred better than the worst case we had budgeted for. With a large, tightly packed field near the top of the leaderboard, small differences in macro $F_{0.5}$ correspond to comparatively large rank movements. The final placement (top 2.9%) improved on the position implied by the provisional score seen earlier in the process, consistent with late movement as the private-leaderboard portion of the test set was scored.

Item	Value
Final leaderboard score	98.4 / 100 (macro $F_{0.5} \approx 0.984$)
Final rank	≈ 930
Participating teams (public report)	$> 32{,}000$
Approximate percentile	top 2.9%

Table 3: Final competition outcome. The team count is drawn from a public participant report, not an organiser statistic, and should be treated accordingly.

9 Limitations

Stated plainly, in roughly decreasing order of concern:

1. **France is unvalidated.** The acceptance threshold and every feature were calibrated on US/India labels only; France transfer is inferred, not measured, and no France-specific normalisation or calibration was possible.
2. **Score provenance.** Earlier reported scores of 0.951/0.952 in our own experiment history could not be independently reproduced from artifacts available at finalisation time, and are noted as unattributed rather than relied upon.
3. **Full-scale operational risk.** No representative full-scale run had, at finalisation time, established memory or runtime behaviour at the full 12.5M-training-row / ∼1.73M-test-S1 scale on the available hardware (16 GiB RAM); the implementation materialises comparatively large in-memory representations.
4. **Evidence level of ablations.** All ablation numbers in Table 2 come from small, sampled screening audits and are not directly comparable across harness versions or to the full-scale result.
5. **Stale artifacts.** Saved model/config artifacts from earlier iterations were schema-incompatible with the final pipeline and were not used to produce the submitted output; a fresh, matching training/inference run was required.

10 Related Work

Probabilistic record linkage traces to Fellegi and Sunter [1], who established that the optimal linkage rule thresholds a likelihood ratio over field agreements; our single-stage classifier with a calibrated global acceptance threshold is a learned, non-linear instance of that same idea. Blocking as a recall/cost trade-off is surveyed by Papadakis et al. [2]; our typed-token retrieval combined with exact-key retrieval is a standard token-blocking scheme within that framework. Feature-based learned matchers such as Magellan [3] and neural matchers such as DeepMatcher [4] and Ditto [5] represent the two dominant families of matching model; we chose a feature-based gradient-boosted approach for its transparency under audit and its fit to the available compute budget. LightGBM [6] is used throughout for pairwise scoring.

11 Conclusion

The largest, most defensible gains in this system came from process rather than model complexity: auditing the ground-truth joins before trusting a one-owner-per-target assumption, correcting a retrieval score that had silently been carrying positional information instead of genuine evidence, and rejecting a second-stage rescorer once cross-fitted, multi-seed evidence showed its gain did not replicate. The result is a deliberately small system — one candidate-generation union, one pairwise scorer, one global threshold with an explicit NULL — that reached a final leaderboard score of 98.4/100, placing in the top 2.9% of a field reported at over 32,000 teams.

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